

PROGRAM – 8

Create Kubernetes Pods using YAML Descriptors

Project Name: program_devops_8

Technologies: Kubernetes, Minikube, YAML, Docker (Node.js image)

Step 1: Ensure Minikube is running

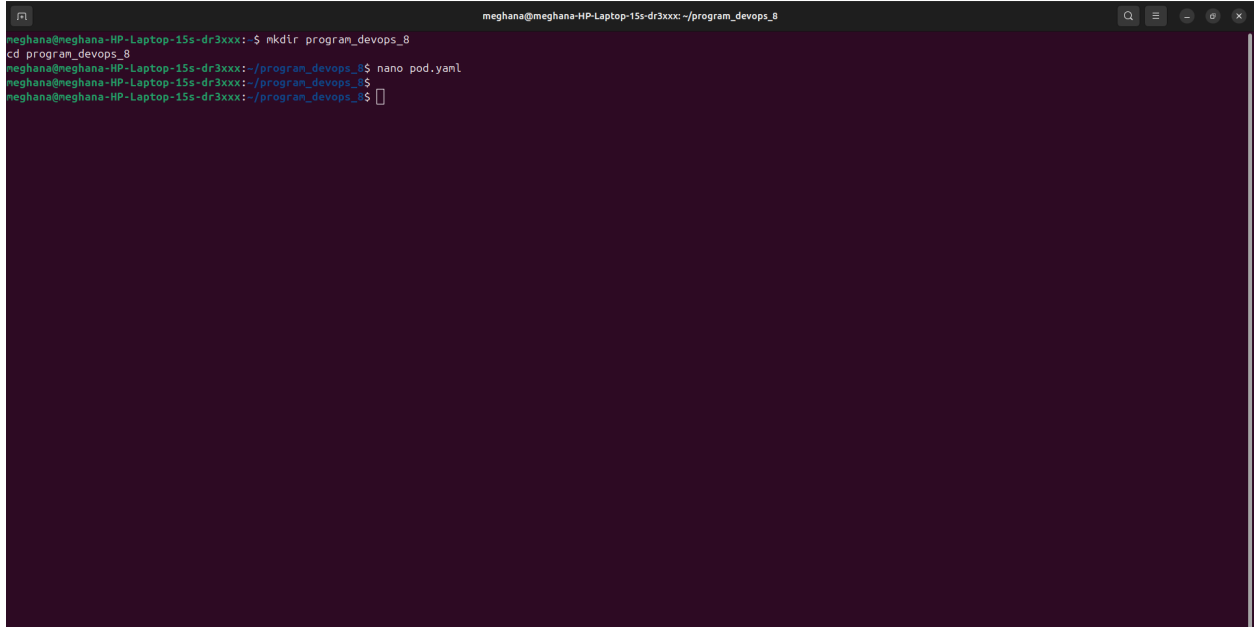
minikube start

Step 2: Verify cluster status

kubectl get nodes

Step 3: Create project directory

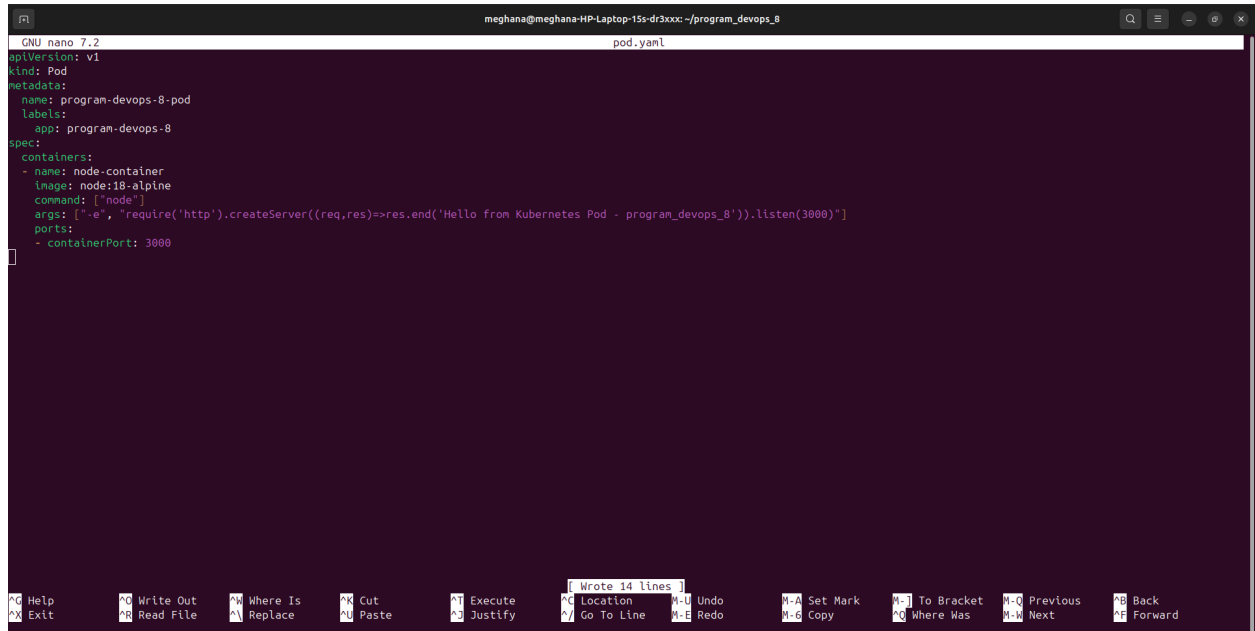
mkdir program_devops_8 cd program_devops_8



```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_8
meghana@meghana-HP-Laptop-15s-dr3xxx:~$ mkdir program_devops_8
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_8$ cd program_devops_8
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_8$ nano pod.yaml
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_8$
```

Step 4: Create Pod YAML descriptor

nano pod.yaml



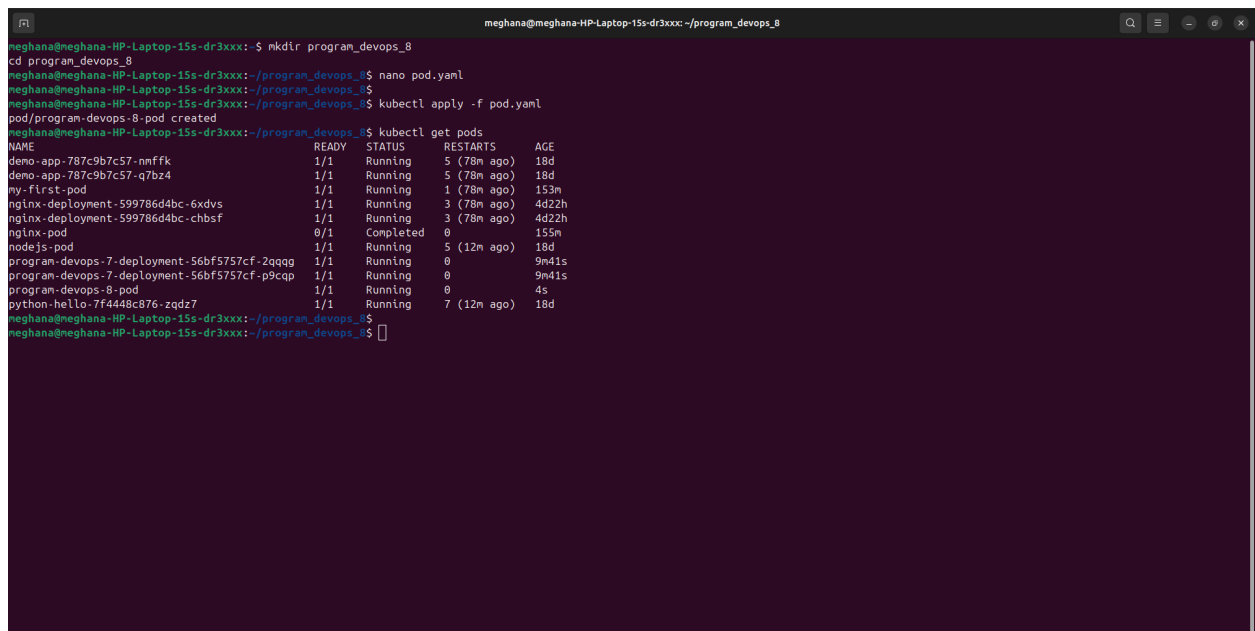
```
GNU nano 7.2 pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-devops-8-pod
  labels:
    app: program-devops-8
spec:
  containers:
  - name: node-container
    image: node:18-alpine
    command: ["node"]
    args: ["-e", "require('http').createServer((req,res)=>res.end('Hello from Kubernetes Pod - program-devops_8')).listen(3000)"]
    ports:
    - containerPort: 3000
```

Step 5: Create Pod using YAML

kubectl apply -f pod.yaml

Step 6: Verify Pod creation

kubectl get pods



```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_8
$ mkdir program_devops_8
$ cd program_devops_8
$ nano pod.yaml
$ kubectl apply -f pod.yaml
pod/program-devops-8-pod created
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
demo-app-787c9b7c57-nmffk           1/1     Running   5 (78m ago)  18d
demo-app-787c9b7c57-q7bz4           1/1     Running   5 (78m ago)  18d
my-first-pod                         1/1     Running   1 (78m ago)  153m
nginx-deployment-599786d4bc-6xdvs    1/1     Running   3 (78m ago)  4d22h
nginx-deployment-599786d4bc-chbsf    1/1     Running   3 (78m ago)  4d22h
nginx-pod                           0/1     Completed 0             155m
nodejs-pod                          1/1     Running   5 (12m ago)  18d
program-devops-7-deployment-56bf5757cf-2qqqg  1/1     Running   0             9m41s
program-devops-7-deployment-56bf5757cf-p9cqg  1/1     Running   0             9m41s
program-devops-8-pod                 1/1     Running   0             4s
python-hello-7f4448c876-zqdz7        1/1     Running   7 (12m ago)  18d
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_8
```

Step 7: Describe Pod

kubectl describe pod program-devops-8-pod

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_8
program-devops-8-pod          1/1      Running    0          4s
python-hello-7f448c876-zqd7z 1/1      Running    7 (12m ago) 18d
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_8$ kubectl describe pod program-devops-8-pod
Name:                         program-devops-8-pod
Namespace:                    default
Priority:                      0
Service Account:              default
Node:                         minikube/192.168.49.2
Start Time:                   Mon, 05 Jan 2026 12:55:00 +0530
Labels:                       app=program-devops-8
Annotations:                  <none>
Status:                       Running
IP:                           10.244.0.79
IPs:                          10.244.0.79
Containers:
  node-container:
    Container ID:   docker://1be843105f717da90dc28d46ff933a2639d288081bf9ffc2c1b096e7b166a776
    Image:          node:18-alpine
    Image ID:       docker-pullable://node@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
    Port:           3000/TCP
    Host Port:      0/TCP
    Command:
      node
    Args:
      -e
      require('http').createServer((req,res)=>res.end('Hello from Kubernetes Pod - program_devops_8')).listen(3000)
    State:          Running
      Started:      Mon, 05 Jan 2026 12:55:01 +0530
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-wxq2x (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                  True
  Ready                        True
  ContainersReady             True
  PodScheduled                 True
```

Step 8: Expose Pod (temporary service)

kubectl expose pod program-devops-8-pod \ --type=NodePort \ --port=3000

Step 9: Get Service details

kubectl get services

```
Normal Started 15s kubelet Started container node-container
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_8$ kubectl expose pod program-devops-8-pod \
--type=NodePort \
--port=3000
service/program-devops-8-pod exposed
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_8$ kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>       443/TCP          18d
nginx-service       NodePort    10.108.163.146 <none>       80:32452/TCP     4d22h
program-devops-7-service NodePort    10.104.241.202 <none>       3000:30007/TCP   9m28s
program-devops-8-pod NodePort    10.102.107.194 <none>       3000:32718/TCP   5s
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_8$ minikube service program-devops-8-pod
```

NAMESPACE	NAME	TARGET PORT	URL
default	program-devops-8-pod	3000	http://192.168.49.2:32718

```
🔗 Opening service default/program-devops-8-pod in default browser...
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_8$ Opening in existing browser session.
```

Step 10: Access Pod application

minikube service program-devops-8-pod

